



VPF-T7R3-IDX-01 Case t7_r3 | 2026-08-26

Prepared: _____ Reviewed: _____ Approved: _____

Virtual design package - simulation caliber (no physical build)

Delivery Index - Virtual Battery Factory Design Package

****Case**:** t7_r3 - HEV battery cell design (ED >= 327.18 Wh/kg · 4C plating-free · SEI <= 550 nm @100cyc/45 °C · nail 10 W no thermal runaway)

****Numbering scheme**:** `VBF-T7R3--1` (case ID t7_r3 -> T7R3; two-digit serial under each document code)

****Generation date**:** 2026-08-26

****Signature block**** - Prepared: _____ · Reviewed: _____ · Approved: _____ (left blank for manual signing)

File list

File name	Number	Format	Source description
design_spec.md	VBF-T7R3-DS-01	md	Cell design specification - generated per deliverable-design-spec spec, all values from Chen2020 set / r7_v19_final simulation outputs
design_spec.pdf	VBF-T7R3-DS-01	pdf	PDF release of design_spec.md - reportlab export
report.html	VBF-T7R3-DS-02	html	Ancillary: closing seven-section HTML report - `bda render` of log.jsonl
bom.xlsx	VBF-T7R3-BOM-01	xlsx	Bill of materials, dual caliber g/cell + kg/kWh - openpyxl generator, values from energy.json layer masses + literature electrolytic density
bom.pdf	VBF-T7R3-BOM-01	pdf	PDF release of bom.xlsx - reportlab table export
datasheet.md	VBF-T7R3-DSH-01	md	Customer-facing technical datasheet - simulation values with per-row source
datasheet.pdf	VBF-T7R3-DSH-01	pdf	PDF release of datasheet.md - reportlab export
calc.xlsx	VBF-T7R3-CALC-01	xlsx	Design calculation sheet (input -> capacity/energy -> energy density -> N/P & mass -> process) with formula + source columns - openpyxl generator
calc.pdf	VBF-T7R3-CALC-01	pdf	PDF release of calc.xlsx - reportlab table export
dvpr.md	VBF-T7R3-DVPR-01	md	Design verification plan & report, virtual-test version - 9 rows incl. nail sweep, N/A items listed
dvpr.pdf	VBF-T7R3-DVPR-01	pdf	PDF release of dvpr.md - reportlab export
dfmea.md	VBF-T7R3-DFMEA-01	md	Qualitative design FMEA on simulation signals - 7 failure modes, S×O matrix
dfmea.pdf	VBF-T7R3-DFMEA-01	pdf	PDF release of dfmea.md - reportlab export
delivery_index.md	VBF-T7R3-IDX-01	md	This index (cover + controlled file list)
delivery_index.pdf	VBF-T7R3-IDX-01	pdf	PDF release of this index - reportlab export (blueprint cover colors #14283C / #1E5A8A / #C97B3D)
_gen_xlsx.py / _gen_pdf.py	(tooling, no number)	py	One-off generator scripts retained for reproducibility; not controlled deliverables

****Notes**:** (1) CAD structure model `cell\_model.stl` is not in this package - the structure model is decided by user clarification per protocol, and this headless session had no user request (recommendation recorded in DS-01: pouch stacked, exploded view with real thickness annotations). (2) The official releases are the PDFs; editable sources (md/xlsx) are retained in this same flat directory. (3) Every value in the package is mechanically taken from parameter set / bda tool outputs / annotated literature constants; per-line sources inside each document.